```
a= [1,2,3,4,5]
print(a)
```

```
[1, 2, 3, 4, 5]
```

```
import numpy as np
arr=np.array([1,2,3,4,5])
print(arr)
```

```
[1 2 3 4 5]
```

```
a=[1,2,3]
b=[4,5,6]
sem=[]
for i in range(len(a)):
    sem.append(a[i]+b[i])
print(sem)
```

```
[5, 7, 9]
```

```
import numpy as np
a=np.array([1,2,3])
b=np.array([4,5,6])
print(a+b)
```

```
[5 7 9]
```

```
import numpy as np
a=np.array([1,2,3])
b=np.array([4,5,6])
print(a+b)
print(a-b)
print(a*b)
print(a**b)
```

```
[5 7 9]
[-3 -3 -3]
[ 4 10 18]
[  1  32 729]
```

```
import numpy as py
arr=np.array([1,2,3,4,5])
ar=np.array([[6,7,8],[9,10,88]])
a=np.array([[[1,2],
            [3,4],
            [5,6]]])
print(arr)
print(ar)
print(a)
print(ar.shape)
```

```
[1 2 3 4 5]
[[ 6  7  8]
 [ 9 10 88]]
[[[1 2]
  [3 4]
  [5 6]]]
(2, 3)
```

```python
import numpy as py
arr=np.array([1,2,3,4,5])
ar=np.array([[6,7,8],[9,10,88]])
a=np.array([[[1,2],
            [3,4],
            [5,6]]])
print(arr)
print(ar)
print(a)
print(ar.shape)
print(arr.ndim)
print(ar.ndim)
print(a.ndim)
```

```
[1 2 3 4 5]
[[ 6  7  8]
 [ 9 10 88]]
[[[1 2]
  [3 4]
  [5 6]]]
(2, 3)
1
2
3
```

```python
import numpy as py
arr=np.array([1,2,3,4,5])
ar=np.array([[6,7,8],[9,10,88]])
a=np.array([[[1,2],
            [3,4],
            [5,6]]])
print(arr)
print(ar)
print(a)
print(ar.shape)
print(arr.ndim)
print(arr.shape)
print(ar.ndim)
print(a.ndim)
print(a.size)
print(ar.size)
print(arr.size)
```

```
print(arr.dtype)
print(a.shape)
```

```
[1 2 3 4 5]
[[ 6  7  8]
 [ 9 10 88]]
[[[1 2]
  [3 4]
  [5 6]]]
(2, 3)
1
(5,)
2
3
6
6
5
int64
(1, 3, 2)
```

```
import numpy as np
a=np.array([1,2,3,4,5])
b=np.zeros(5)
print(a)
print(b)
c=np.zeros((2,2))
print(c)
d=np.ones(3)
print(d)
r=np.ones(3)
```

```
[1 2 3 4 5]
[0. 0. 0. 0. 0.]
[[0. 0.]
 [0. 0.]]
[1. 1. 1.]
[[1. 1.]
 [1. 1.]]
```

```
import numpy as np
a=np.array([1,2,3,4,5])
b=np.zeros(5)
print(a)
print(b)
print(np.arange(1,10,2))
print(np.linspace(0,20,3))
print(np.random.rand(5))
print(np.random.randint(1,100,7))
print(np.random.randn(5))
```

```
[1 2 3 4 5]
[0. 0. 0. 0. 0.]
[1 3 5 7 9]
[ 0. 10. 20.]
[0.27276784 0.99433698 0.24965386 0.61438542 0.13539973]
[43 60 89 24 41 78 16]
[ 0.10455568  0.75601194  0.2715573  -0.55334469  1.79869057]
```

numpy indexing ans slicing

```
import numpy as np
a=np.array([10,20,30,40,50,60,67])
print(a[3])
b=np.array([[1,2,7],
           [3,4,8],
           [5,6,9]])
print(b[1][1])
print(a[2:5])
print(a[:4])
print(a[::2])
print(b[1])
print(b[:,1])
print(a[-1::-3])
print(a[a<60])
print(a[[0,4,6,3]])
print(a[a%2==0])
a[4]=1000
print(a)
```

```
40
4
[30 40 50]
[10 20 30 40]
[10 30 50 67]
[3 4 8]
[2 4 6]
[67 40 10]
[10 20 30 40 50]
[10 50 67 40]
[10 20 30 40 50 60]
[  10   20   30   40 1000   60   67]
```

mathematical operation in numpy

```
import numpy as np
a=np.array([1,2,3])
```

```
print(a+10)
b=np.array([[1,2,3],
            [4,5,6]])
print(b+10)
print(b+a)
```

```
[11 12 13]
[[11 12 13]
 [14 15 16]]
[[2 4 6]
 [5 7 9]]
```

matrix operation

```
import numpy as py
a=np.array([[1,2],
            [3,4]])
b=np.array([[4,3],
            [2,1]])
print(a*b)
print(np.dot(a,b))
print(a@b)
print(np.matmul(a,b))
print(a.T)# convert row into column
```

```
[[4 6]
 [6 4]]
[[ 8  5]
 [20 13]]
[[ 8  5]
 [20 13]]
[[ 8  5]
 [20 13]]
[[1 3]
 [2 4]]
```

universal mathematical function

```
import numpy as py
a=np.array([4,9,16])
print(np.sqrt(a))
b=np.array([-5,-6,-2,-8])
print(np.abs(b))
print(np.log(a))
print(np.exp(a))
```

```
[2. 3. 4.]
[5 6 2 8]
[1.38629436 2.19722458 2.77258872]
[5.45981500e+01 8.10308393e+03 8.88611052e+06]
```

```
#round function
arr=np.array([1.234,5.674])
print(np.round(arr,2))
print(np.round(arr,3))
```

```
[1.23 5.67]
[1.234 5.674]
```

statistical function

```
import numpy as np
a=np.array([10,2,3,40,50])
print(np.mean(a))
print(np.median(a))
print(np.std(a))
print(np.max(a))
print(np.min(a))
print(np.sum(a))
print(np.percentile(a,25))
print(np.percentile(a,50))
print(np.percentile(a,75))
print(np.var(a))
print(np.argmin(a))
print(np.argmax(a))
b=np.array([1,2,3,4,5])
c=np.array([2,4,6,8,10])
print(np.corrcoef(b,c))
d=np.array([[1,3,4],
           [5,6,8]])
print(np.sum(d))
print(np.sum(d,axis=1))
print(np.sum(d,axis=0))
```

```
21.0
10.0
20.039960079800558
50
2
105
3.0
10.0
40.0
401.6
1
4
[[1. 1.]
 [1. 1.]]
27
```

```
[ 8 19]
[ 6  9 12]
```

Workout question

```
# Create 2D array  of student scores,complete class average,highest score, normalized score to 0 to 1

del print

import numpy as np
a = np.array([
    [50, 60, 80, 90],
    [70, 90, 60, 50],
    [90, 50, 40, 30]
])
score = np.sum(a, axis=1)
subavg=(np.sum(a,axis=0))
print(score)
print(subavg)
print(np.mean(a))
print(np.max(subavg))
```

```
[280 270 210]
[210 200 180 170]
63.333333333333336
210
```

```
import numpy as np
arr = np.array([10,20,30,40,50])
print(arr[arr>25])
print(sum(arr[arr>25]))
```

```
[30 40 50]
120
```

```
import numpy as np
a = np.array([[3,1],
             [1,4]])
print(a.shape)
```

```
(2, 2)
```